

Abstract

In the highly competitive optical retail market, Specsmaakers strives to optimize its operational processes while meeting customer expectations. With over 250 stores spread across five states, Specsmaakers must manage extensive data to achieve customer satisfaction and generate healthy profits. Big data analytics is utilized at every stage of the business process in retail markets.

This project focuses on analyzing Turnaround Time (TAT) for various processes involved in job order lifecycles at Specsmaakers using Power BI. The analysis encompasses the entire workflow from order placement to final product dispatch, aiming to identify bottlenecks and improve operational efficiency. By leveraging data from the Optic Pro application, which is extracted in CSV format, and using tables such as Store Master, Sales Register, Date Table, and Lens Master, the project demonstrates how Power BI can be used to create interactive dashboards that visualize TAT metrics.

The project illustrates the application of business analytics in shaping effective operational strategies through data sources, data structures, and DAX query language. It provides insights into historical trends, business performance, and critical operational metrics, including order processing times, stock management efficiency, vendor performance, and fitting processes. Additionally, the project includes a "what-if" analysis for future planning based on historical trends. This dashboard is designed for stakeholders to understand the processes, identify functional areas for improvement, and visualize performance metrics across various dimensions.

This project represents a comprehensive approach to transforming large datasets into actionable insights, enabling stakeholders to quickly assess performance and make informed decisions.

1. Introduction

1.1. Power BI

Power BI is a powerful business analytics tool developed by Microsoft that enables users to visualize and share insights from their data. It offers interactive dashboards, robust reporting capabilities, and a user-friendly interface that allows businesses to turn raw data into actionable insights. Power BI integrates seamlessly with various data sources, allowing for comprehensive data analysis and visualization. Key features include:

Interactive Dashboards: Create visually appealing and interactive dashboards that provide a real-time view of business performance.

Data Connectivity: Connect to a wide range of data sources, including databases, cloud services, and flat files.

Advanced Analytics: Utilize built-in advanced analytics capabilities, such as DAX (Data Analysis Expressions) for complex calculations.

Collaboration: Share reports and dashboards with team members and stakeholders to facilitate collaborative decision-making.

Mobile Access: Access reports and dashboards on mobile devices, ensuring insights are available anytime, anywhere.

1.2. TAT Analysis

Turnaround Time (TAT) analysis involves the measurement and analysis of the time taken to complete specific processes from start to finish. In the context of Specs-makers, TAT analytics focuses on the entire job order lifecycle, from when an order is placed to the final product dispatch. By analyzing TAT, businesses can identify inefficiencies, optimize workflows, and enhance overall operational performance. Specific benefits of TAT analytics include:

Performance Monitoring: Track and monitor the performance of different stages in the order fulfillment process.

Benchmarking: Establish benchmarks for acceptable turnaround times and compare actual performance against these benchmarks.

Root Cause Analysis: Identify the root causes of delays and implement corrective actions to improve process efficiency.

Predictive Insights: Use historical TAT data to predict future performance and plan accordingly.

1.3. Why Should Retailers Care About TAT Analysis

Retailers should care about TAT analysis because it directly impacts customer satisfaction and operational efficiency. By reducing turnaround times, retailers can achieve several key advantages:

Improved Customer Satisfaction: Faster order fulfillment leads to happier customers and increased loyalty.

Operational Efficiency: Streamlined processes reduce waste and increase productivity, leading to cost savings.

Competitive Advantage: Efficient operations and quick service can differentiate a retailer from competitors, attracting more customers.

Inventory Management: Better alignment of stock levels with demand reduces overstock and stockouts.

Profitability: Optimized processes and reduced costs contribute to healthier profit margins.

In a competitive market, faster and more efficient service can be a significant differentiator, leading to increased customer loyalty and business growth.

1.4. What is TAT Data Analysis

TAT data analysis is the process of collecting, processing, and analyzing data related to turnaround times in order to gain insights and improve performance. This involves several key steps:

Data Collection: Gathering data from various sources, such as order management systems, inventory records, and sales transactions.

Data Processing: Cleaning and organizing data to ensure accuracy and consistency.

Data Analysis: Using statistical and analytical tools to identify patterns, trends, and areas for improvement.

Visualization: Creating dashboards and reports that present data in an easy-to-understand format, enabling stakeholders to make informed decisions.

Continuous Improvement: Utilizing insights from TAT data to drive continuous improvement initiatives across the organization. For Specsmakers, TAT data analytics provides a comprehensive view of the operational efficiency across different stages of the job order lifecycle, helping to identify bottlenecks and areas for improvement.

1.5. System Requirements

To effectively implement and utilize Power BI for TAT analysis, the following system requirements should be met:

Hardware Requirements:

Processor: At least a 1.6 GHz or faster x64 processor.

Memory: 4 GB of RAM (8 GB or more recommended for large datasets).

Storage: A minimum of 1 GB available hard disk space for installation (additional space required for data storage).

Display: 1440 x 900 resolution or higher.

Software Requirements:

Operating System: Windows 7 / Windows Server 2008 R2, or later versions.

.NET Framework: Version 4.6 or later.

Web Browser: Microsoft Edge, Internet Explorer 11, or the latest versions of Chrome or Firefox.

Additional Requirements:

Data Sources: Access to data from the Optic Pro application in CSV format.

Network: Reliable internet connection for data synchronization and cloud services.

Licensing: Appropriate Power BI licensing, such as Power BI Pro or Power BI Premium, depending on organizational needs.

Integration Tools: Tools for integrating various data sources into Power BI, such as Power Query.

User Training: Training sessions for users to understand how to navigate and utilize Power BI dashboards effectively.

Security Protocols: Implement security measures to protect sensitive data and ensure compliance with data protection regulations.

Meeting these requirements ensures that Specsmakers can leverage Power BI effectively to perform TAT analytics, leading to better insights and improved operational performance.

2. Background

2.1. Scalability and Performance

Scalability and performance are crucial considerations for any analysis platform, especially in the context of Specsmakers, which operates over 250 stores across five states. Power BI offers scalability by allowing users to seamlessly scale up or down based on their business needs. It can handle large volumes of data and complex calculations efficiently, ensuring smooth performance even as data volumes grow. Additionally, Power BI's cloud-based architecture enables Specsmakers to leverage the scalability and

resources of the Microsoft Azure platform, further enhancing performance and ensuring reliable access to analysis insights.

Name	Duration (ms) ↓
🕒 <i>Recording started (5/30/2024 4:05:51 PM)</i>	-
⊕ Slicer	69
⊕ Slicer	28
⊕ Slicer	53
⊕ Slicer	34
⊕ Slicer	52
⊕ Slicer	50
🖨️ <i>Cross-highlighted</i>	-
⊕ Slicer	183
⊕ Slicer	182
⊕ Slicer	181
⊕ Slicer	181
⊕ Slicer	180

Figure 01. Referring performance analyzer if I set 1 Lacs records

Duration calculated in microseconds it will take hardly 1 min data and all visualization refreshes.

2.2. Accessibility

Accessibility is key to ensuring that stakeholders across Specsmakers' organization can access and utilize analysis insights effectively. Power BI offers a range of accessibility features, including:

Web-Based Access: Users can access Power BI dashboards and reports via a web browser, enabling easy access from any device with an internet connection.

Mobile Apps: Power BI provides mobile apps for iOS, Android, and Windows devices, allowing users to view and interact with dashboards on the go.

Sharing and Collaboration: Power BI enables users to share dashboards and reports with colleagues, partners, and customers, facilitating collaboration and decision-making.

Embedding: Power BI dashboards and reports can be embedded into custom applications or websites, extending accessibility to external stakeholders and customers.

By providing multiple avenues for accessing analytics insights, Power BI ensures that stakeholders across Specsmakers' organization can make data-driven decisions wherever they are and whenever they need to.

2.3. GUI (Graphical User Interface)

Power BI's intuitive graphical user interface (GUI) makes it easy for users to create, customize, and interact with dashboards and reports. Key features of Power BI's GUI include:

Drag-and-Drop Interface: Users can easily drag and drop data fields to create visualizations, eliminating the need for complex coding or scripting.

Interactive Visualizations: Power BI offers a wide range of interactive visualization options, such as charts, graphs, maps, and tables, that users can customize and interact with to gain insights.

Formatting Options: Users can customize the appearance of visualizations, including colors, fonts, and labels, to ensure that dashboards and reports are visually appealing and easy to understand.

Natural Language Queries: Power BI's natural language query feature allows users to ask questions about their data using everyday language, making it easier to explore and analyze data without requiring technical expertise.

Overall, Power BI's GUI empowers users across Specsmakers' organization to create and interact with analytics insights in a user-friendly and intuitive manner, driving better decision-making and business outcomes.

3. Methodology

The main objective of creating this dashboard is to enable stakeholders to directly analyze the Turnaround Time (TAT) for various processes involved in job order lifecycles at Specsmakers. This includes identifying areas where delays occur, such as order processing, stock management, and fitting processes. Additionally, a "what-if" analysis feature is included to evaluate the impact of changes in key parameters, such as a 5% increase or decrease in processing speed, on overall TAT. This functionality aids decision-makers in identifying bottlenecks, improving operational efficiency, and understanding business performance comprehensively.

3.2. Data Source

TAT raw data taken from Optic Pro

There are 4 Tables,

Timing Report:-

Below are the Columns with description:

City: The city where the store is located.

Location: The specific location or neighborhood of the store within the city.

Store Code: A unique identifier for each store.

Lab Location: The location of the lab where the orders are processed.

Indent Number: A unique identifier for the indent (internal request) within the system.

Order Number: A unique identifier for each order placed.

Order Date: The date when the order was placed.

Order Type: The type of order (e.g. Complete Spectacles, Contact Lens).

Status: The current status of the order (e.g., received, in process, completed, dispatched).

Status Date: The date when the current status was last updated.

PairCode: A code representing a specific pair of lenses or glasses associated with the order. (Primary Key)

SNO	City	Location	Store Code	Lab Location	Indent Number	Order Number	Order Date	Order Type	Status	Status Date	PairCode
1	CHENNAI	Store 1	ST001	Lab Chennai	INL01-000001	ST001-AB000001	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 17:44	ST001-AB000001-1
2	CHENNAI	Store 1	ST001	Lab Chennai	INL01-000002	ST001-AB000002	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 17:44	ST001-AB000002-2
3	CHENNAI	Store 2	ST002	Lab Chennai	INL01-000003	ST002-AB000003	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 14:49	ST002-AB000003-1
4	CHENNAI	Store 2	ST002	Lab Chennai	INL01-000004	ST002-AB000004	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 14:49	ST002-AB000004-2
5	CHENNAI	Store 2	ST002	Lab Chennai	INL01-000005	ST002-AB000005	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 20:19	ST002-AB000005-1
6	CHENNAI	Store 2	ST002	Lab Chennai	INL01-000006	ST002-AB000002	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 20:19	ST002-AB000002-2
7	CHENNAI	Store 3	ST003	Lab Chennai	INL01-000007	ST003-AB000001	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 13:49	ST003-AB000001-1
8	CHENNAI	Store 3	ST003	Lab Chennai	INL01-000008	ST003-AB000008	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 16:36	ST003-AB000008-2
9	CHENNAI	Store 3	ST003	Lab Chennai	INL01-000009	ST003-AB000001	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 16:36	ST003-AB000001-1
10	CHENNAI	Store 3	ST003	Lab Chennai	INL01-000010	ST003-AB000002	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 16:41	ST003-AB000002-2
11	CHENNAI	Store 3	ST003	Lab Chennai	INL01-000011	ST003-AB000001	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 16:41	ST003-AB000001-1
12	CHENNAI	Store 3	ST003	Lab Chennai	INL01-000012	ST003-AB0000012	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 18:36	ST003-AB0000012-2
13	CHENNAI	Store 3	ST003	Lab Chennai	INL01-000013	ST003-AB000001	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 19:01	ST003-AB000001-1
14	CHENNAI	Store 3	ST003	Lab Chennai	INL01-000014	ST003-AB000002	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 19:01	ST003-AB000002-2
15	CHENNAI	Store 4	ST004	Lab Chennai	INL01-000015	ST004-AB000001	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 17:24	ST004-AB000001-1
16	CHENNAI	Store 4	ST004	Lab Chennai	INL01-000016	ST004-AB000002	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 17:24	ST004-AB000002-2
17	CHENNAI	Store 5	ST005	Lab Chennai	INL01-000017	ST005-AB000001	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 14:49	ST005-AB000001-1
18	CHENNAI	Store 5	ST005	Lab Chennai	INL01-000018	ST005-AB000002	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 14:49	ST005-AB000002-2
19	CHENNAI	Store 5	ST005	Lab Chennai	INL01-000019	ST005-AB000001	25-May-24	SPECTACLE	Job Order Raised In Store	25-05-2024 16:41	ST005-AB000001-1

Figure 02.Timing Report

Sales Register:-

Below are the Columns with description:

Date: The date of a transaction or event.

Invoice No: Unique identifier for each invoice.

Barcode: Unique code for product identification.

Store Code: Unique identifier for each store.

PairCode: Code representing a specific pair of lenses or glasses.

City Master:-

Below are the Columns with description:

Store Code: Unique identifier for each store.

Store Name: Name of the store.

City: The city where the store is located.

City Group: Grouping of cities for categorization or analysis purposes.

Store_Code	Store_Name	City_Group	City_Name
AB001	Store 1	CHENNAI	Chennai
AB002	Store 2	CHENNAI	Chennai
AB003	Store 3	CHENNAI	Chennai
AB004	Store 4	CHENNAI	Chennai
AB005	Store 5	CHENNAI	Chennai
AB006	Store 6	CHENNAI	Chennai
AB007	Store 7	CHENNAI	Chennai
AB008	Store 8	CHENNAI	Chennai
AB009	Store 9	CHENNAI	Chennai
AB010	Store 10	CHENNAI	Chennai
AB011	Store 11	CHENNAI	Chennai
AB012	Store 12	CHENNAI	Chennai
AB013	Store 13	CHENNAI	Chennai
AB014	Store 14	CHENNAI	Chennai
AB015	Store 15	CHENNAI	Chennai
AB016	Store 16	CHENNAI	Chennai
AB017	Store 17	CHENNAI	Chennai
AB018	Store 18	CHENNAI	Chennai
AB019	 re 19	CHENNAI	Chennai

Figure 03.City Master

Lens wise TAT Master:-

Below are the Columns with description:

Barcode: Unique code for product identification.

Product Group: Category or group to which the product belongs.

Brand: The brand name of the product.

Chennai TAT: Turnaround Time for orders processed in Chennai.

Bangalore TAT: Turnaround Time for orders processed in Bangalore.

ROTN TAT: Turnaround Time for orders processed in the Rest of Tamil Nadu.

ROKN TAT: Turnaround Time for orders processed in the Rest of Karnataka.

AP&TS TAT: Turnaround Time for orders processed in Andhra Pradesh and Telangana.

Product Group1	BRANDS	Lens Group (Index)	COATINGS	Barcode	CHE	ROTN	TS & AP CUT LENS	TS & AP FITT	BLR	ROKN
BI FOCAL	EVA		1.5 ARC	###AB12	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB13	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB14	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB15	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB16	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB17	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB18	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB19	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB20	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB21	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB22	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB23	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB24	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB25	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB26	5	6	6	6	5	6
BI FOCAL	EVA		1.5 ARC	###AB27	5	6	6	6	5	6
BI FOCAL	EVA		1.5 BLU	###AB28	5	6	6	6	5	6
BI FOCAL	EVA		1.5 BLU	###AB29	5	6	6	6	5	6
BI FOCAL	EVA		1.5 BLU	###AB30	5	6	6	6	5	6

Figure 04 . Excel raw data

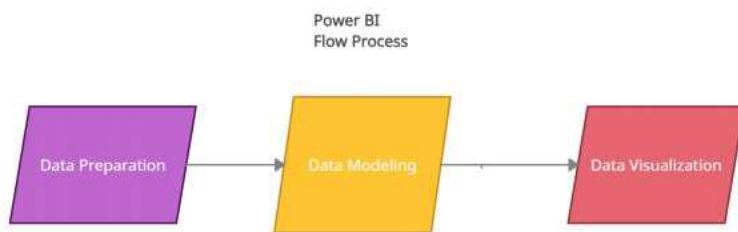


Figure 05.Power BI Flow Process

1. Data preparation – organizing the data in a flat table.

2. Data modeling – It is used to connect multiple data sources in BI tool using a relationship.

3. Data Visualization – It is the process of transforming huge data sets into graphs charts and other visualizations.

3.3. How to upload raw data into Power BI

To Create Power BI, visualize dashboard, you need to download Power bi desktop and install.

Once you have installed, then we need to follow below process (Microsoft Power BI documentation, 2021):

Click on Get Data => Excel workbook = > browse your raw file = > select required sheets and then click on transform => Power Query editor window will be open => then from each table you have to check all data types, then you have to click on close & apply.

Now your base data is ready!

Date Table:

I have created date table for measures and apply into charts. Below is DAX query:

DateTable =

```

ADDCOLUMNS (
    CALENDAR (MIN('summarize table'[Order Date]), MAX('summarize table'[Order Date])),
    "Year", YEAR([Date]),
    "Quarter", QUARTER([Date]),
    "Month", FORMAT([Date], "MM"),
    "Week", INT(( [Date] - DATE(YEAR([Date]), 1, 2) - WEEKDAY(DATE(YEAR([Date]), 1, 2), 2) + 10) / 7),
    "Week Start Date", [Date] - WEEKDAY([Date], 2) + 1, // Calculate Monday as the start of the week
    "Week End Date", [Date] - WEEKDAY([Date], 2) + 7, // Calculate Sunday as the end of the week
    "Custom Week Number", CONCATENATE("Week - ", INT(( [Date] - DATE(YEAR([Date]), 1, 2) - WEEKDAY(DATE(YEAR([Date]), 1, 2), 2) + 10) / 7))
)

```

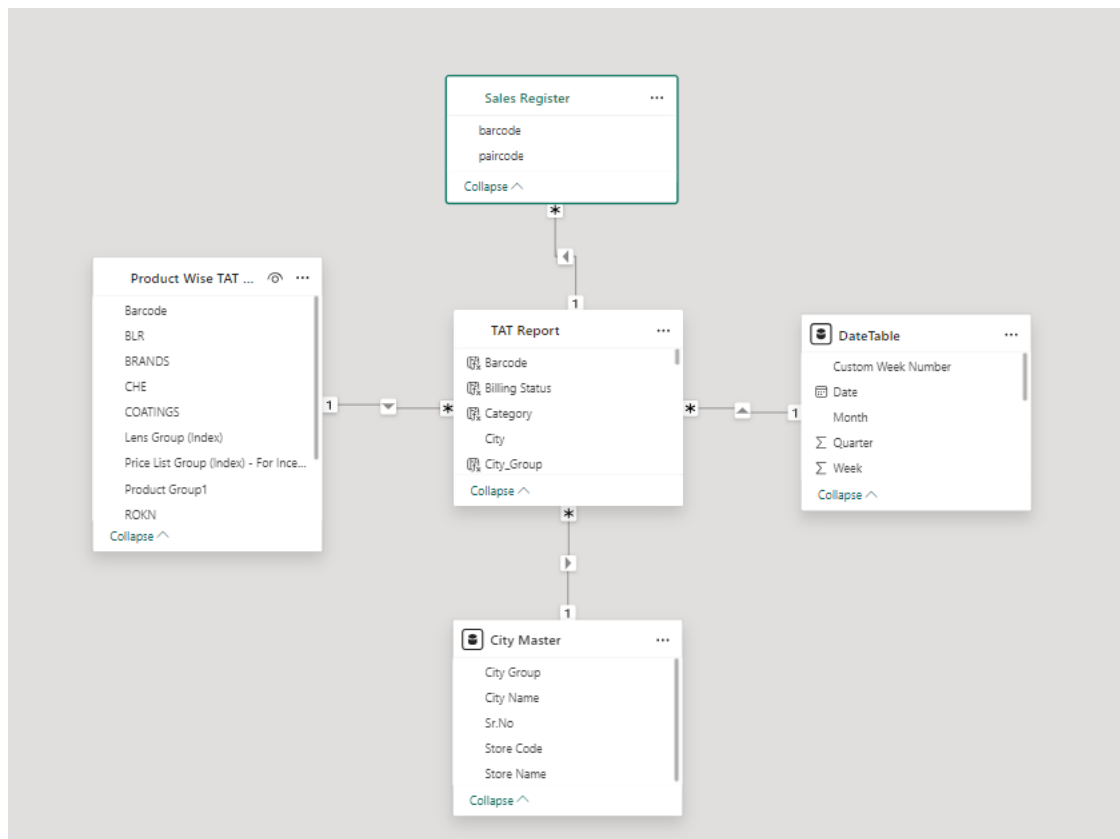


Figure 06. Connection between multiple tables for calculations

3.4. Data Analysis Expressions (DAX)

Data Analysis Expressions (DAX) is a query programming language that is used throughout Microsoft Power BI for creating calculated columns, measures, and custom tables. It is a collection of functions, operators, and constants that can be used in a formula, or expression, to

calculate and return one or more values.

3.5. DAX Queries

I have created multiple measures with DAX query:

```
1)TAT Report = SUMMARIZECOLUMNS(  
    'Timing Report Oct-Mar'[City],  
    'Timing Report Oct-Mar'[Indent Number],  
    'Timing Report Oct-Mar'[Location],  
    'Timing Report Oct-Mar'[Order Date],  
    'Timing Report Oct-Mar'[PairCode],  
    'Timing Report Oct-Mar'[Order Type],  
    'Timing Report Oct-Mar'[Store Code],  
    'Timing Report Oct-Mar'[Lab Location],  
    "Job Order Raised", CALCULATE(FIRSTNONBLANK('Timing Report Oct-Mar'[Status Date], 1), FILTER('Timing  
Report Oct-Mar', 'Timing Report Oct-Mar'[Status] IN {"Job Order Raised In Store"})),  
    "Frame Received with Good Condition", CALCULATE(FIRSTNONBLANK('Timing Report Oct-Mar'[Status  
Date], 1), FILTER('Timing Report Oct-Mar', 'Timing Report Oct-Mar'[Status] IN {"Frame Received with Good  
Condition"})),  
    "Fitment In-House", CALCULATE(FIRSTNONBLANK('Timing Report Oct-Mar'[Status Date], 1), FILTER('Timing  
Report Oct-Mar', 'Timing Report Oct-Mar'[Status] IN {"Fitment In-House"})),  
    "Items Dispatched To Outlet", CALCULATE(FIRSTNONBLANK('Timing Report Oct-Mar'[Status Date], 1),  
FILTER('Timing Report Oct-Mar', 'Timing Report Oct-Mar'[Status] IN {"Items Dispatched To Outlet"})),  
    "Cut Lens", CALCULATE(FIRSTNONBLANK('Timing Report Oct-Mar'[Status Date], 1), FILTER('Timing Report  
Oct-Mar', 'Timing Report Oct-Mar'[Status] IN {"Cut Lens No"})),  
    "Indent Cancel", CALCULATE(FIRSTNONBLANK('Timing Report Oct-Mar'[Status Date], 1), FILTER('Timing  
Report Oct-Mar', 'Timing Report Oct-Mar'[Status] IN {"Indent Cancel", "Fitment Outside"}))
```

This formula summarizes the TAT data by various dimensions (City, Indent Number, etc.) and calculates the first occurrence of specific statuses (e.g., "Job Order Raised", "Frame Received with Good Condition") for each combination of these dimensions.

```
2)Formatted Job Order Date =  
    IF( OR(HOUR('TAT Report'[Job Order Raised]) >= 19,  
WEEKDAY('TAT Report'[Job Order Raised], 2) = 7 // Sunday ),  
    'TAT Report'[Job Order Raised] + 1,'TAT Report'[Job Order Raised])
```

This formula adjusts the "Job Order Raised" date by adding a day if the order was raised after 7 PM or on a Sunday, to reflect the next working day.

3)Job order Raised & Fitment In-House =

```
VAR DateDifference = DATEDIFF('TAT Report'[Formatted JO Date], 'TAT Report'[Fitment In-House], DAY)
RETURN
IF(ISBLANK([Fitment In-House]), 0, IF(DateDifference = 0 OR DateDifference = -1, 1, DateDifference))
```

This calculates the number of days between the adjusted "Job Order Raised" date and the "Fitment In-House" date. If "Fitment In-House" is blank, it returns 0; if the difference is 0 or -1, it returns 1; otherwise, it returns the actual day difference.

4)Status Of Fitment Done =

```
SWITCH(
    TRUE(),
    'TAT Report'[JO& Fitment In-House] >= 20, "6) More than 20 Days",
    'TAT Report'[JO& Fitment In-House] >= 11, "5) 11 - 20 Days",
    'TAT Report'[JO& Fitment In-House] >= 7, "4) 7 - 10 Days",
    'TAT Report'[JO& Fitment In-House] >= 3, "3) 5 - 6 Days",
    'TAT Report'[JO& Fitment In-House] >= 2, "2) 2 - 4 Days",
    'TAT Report'[JO& Fitment In-House] = 0, "7) Fitment not done",
    "1) Less than 2 Days"
)
```

This formula categorizes the time taken from "Job Order Raised" to "Fitment In-House" into different ranges, assigning a descriptive status based on the number of days.

5)Job order Raised to Item Dispatched Outlet =

```
VAR DateDifference = DATEDIFF('TAT Report'[Formatted JO Date], 'TAT Report'[Items Dispatched To Outlet], DAY)
RETURN
IF(ISBLANK([Items Dispatched To Outlet]), 0, IF(DateDifference = 0 OR DateDifference = -1, 1, DateDifference))
```

This calculates the number of days between the adjusted "Job Order Raised" date and the "Items Dispatched To Outlet" date. If "Items Dispatched To Outlet" is blank, it returns 0; if the difference is 0 or -1, it returns 1; otherwise, it returns the actual day difference.

6)Status Of Item Dispatched Outlet =

```
SWITCH(  
    TRUE(),  
    'TAT Report'[JO&IDO] >= 20, "6) More than 20 Days",'TAT Report'[JO&IDO] >= 16, "5) 16 - 20 Days",  
    'TAT Report'[JO&IDO] >= 11, "4) 11 - 15 Days", 'TAT Report'[JO&IDO] >= 6, "3) 6 - 10 Days",  
    'TAT Report'[JO&IDO] >= 2, "2) 2 - 5 Days",'TAT Report'[JO&IDO] = 0, "7) Not Dispatched",  
    "1) Less than 2 Days")
```

This formula categorizes the time taken from "Job Order Raised" to "Items Dispatched To Outlet" into different ranges, assigning a descriptive status based on the number of days.

7)Category =

```
SWITCH (  
    TRUE (),  
    CONTAINSSTRING(UPPER('TAT Report'[PairCode]), "RW"), "RW",  
    CONTAINSSTRING(UPPER('TAT Report'[PairCode]), "LR"), "LR",  
    CONTAINSSTRING(UPPER('TAT Report'[PairCode]), "FTC"), "FTC",  
    CONTAINSSTRING(UPPER('TAT Report'[PairCode]), "FL"), "FL",  
    CONTAINSSTRING(UPPER('TAT Report'[PairCode]), "RR"), "RR",  
    "NORMAL"  
)
```

This formula assigns a category based on the content of the PairCode column. It checks for specific strings (e.g., "RW", "LR") and returns the corresponding category. If none match, it returns "NORMAL".

8)Barcode =

```
CALCULATE(  
    FIRSTNONBLANK('Sales Register'[barcode], 1),  
    FILTER('Sales Register', 'Sales Register'[paircode] = 'TAT Report'[PairCode]))
```

This formula retrieves the first non-blank barcode from the Sales Register table where the paircode matches the PairCode in the TAT Report table.

```
9)City_Group = IF(LOOKUPVALUE('City Master'[City Group],'City Master'[Store Code],'TAT Report'[Store Code]) = "",LOOKUPVALUE('Master Data'[City Group],'Master Data'[Store Code],'TAT Report'[Store Code]),LOOKUPVALUE('City Master'[City Group],'City Master'[Store Code],'TAT Report'[Store Code]))
```

This formula determines the city group by first looking up the city group in the City Master table. If not found, it falls back to looking up the city group in the Master Data table.

```

10)Item Dispatched TAT =
SWITCH(
    TRUE,
    'TAT Report'[City_Group] = "Chennai", IF(ISBLANK(LOOKUPVALUE('Product Wise TAT Master'[CHE], 'Product
Wise TAT Master'[Barcode], 'TAT Report'[Barcode])), 5, LOOKUPVALUE('Product Wise TAT Master'[CHE],
'Product Wise TAT Master'[Barcode], 'TAT Report'[Barcode])),
    'TAT Report'[City_Group] = "ROTN", IF(ISBLANK(LOOKUPVALUE('Product Wise TAT Master'[ROTN], 'Product
Wise TAT Master'[Barcode], 'TAT Report'[Barcode])), 5, LOOKUPVALUE('Product Wise TAT Master'[ROTN],
'Product Wise TAT Master'[Barcode], 'TAT Report'[Barcode])),
    'TAT Report'[City_Group] = "ROKN", IF(ISBLANK(LOOKUPVALUE('Product Wise TAT Master'[ROKN], 'Product
Wise TAT Master'[Barcode], 'TAT Report'[Barcode])), 5, LOOKUPVALUE('Product Wise TAT Master'[ROKN],
'Product Wise TAT Master'[Barcode], 'TAT Report'[Barcode])),
    'TAT Report'[City_Group] = "Bangalore", IF(ISBLANK(LOOKUPVALUE('Product Wise TAT Master'[BLR],
'Product Wise TAT Master'[Barcode], 'TAT Report'[Barcode])), 5, LOOKUPVALUE('Product Wise TAT
Master'[BLR], 'Product Wise TAT Master'[Barcode], 'TAT Report'[Barcode])),
    'TAT Report'[City_Group] = "Hyderabad", IF(ISBLANK(LOOKUPVALUE('Product Wise TAT Master'[TS & AP
FITT], 'Product Wise TAT Master'[Barcode], 'TAT Report'[Barcode])), 5, LOOKUPVALUE('Product Wise TAT
Master'[TS & AP FITT], 'Product Wise TAT Master'[Barcode], 'TAT Report'[Barcode])),
    'TAT Report'[City_Group] = "ROAP", IF(ISBLANK(LOOKUPVALUE('Product Wise TAT Master'[TS & AP FITT],
'Product Wise TAT Master'[Barcode], 'TAT Report'[Barcode])), 5, LOOKUPVALUE('Product Wise TAT Master'[TS
& AP FITT], 'Product Wise TAT Master'[Barcode], 'TAT Report'[Barcode])),
    5)

```

This formula calculates the item dispatched TAT (Turnaround Time) based on the city group. It looks up specific values from the Product Wise TAT Master table, defaulting to 5 if no specific TAT value is found.

```

11)Fitment Achievement = if('TAT Report'[FT TAT]>='TAT Report'[JO& Fitment In-House] && 'TAT Report'[JO&
Fitment In-House]<>0,"Achieved","Not Achieved")

```

This formula determines whether the fitment target has been achieved by comparing the fitment TAT (FT TAT) with the actual TAT from job order raised to fitment in-house (JO& Fitment In-House). It returns "Achieved" if the target is met and "Not Achieved" otherwise.

```

12)Item Dispatched Achievement = if('TAT Report'[IDOTAT]>='TAT Report'[JO&IDO] && 'TAT
Report'[JO&IDO]<>0,"Achieved","Not Achieved")

```

This formula determines whether the item dispatched target has been achieved by comparing the item dispatched TAT (IDOTAT) with the actual TAT from job order raised to item dispatched outlet (JO&IDO). It returns "Achieved" if the target is met and "Not Achieved" otherwise.

13)Billing Status = IF('TAT Report'[Barcode] = "", "Unbilled", Billed")

This formula sets the billing status to "Unbilled" if the barcode is blank and "Billed" otherwise.

14)Fitment Done TAT = 'TAT Report'[Item Dispatched TAT]-1

This formula calculates the fitment done TAT by subtracting 1 from the item dispatched TAT. This provides the TAT value adjusted for the fitment process.

4. Results

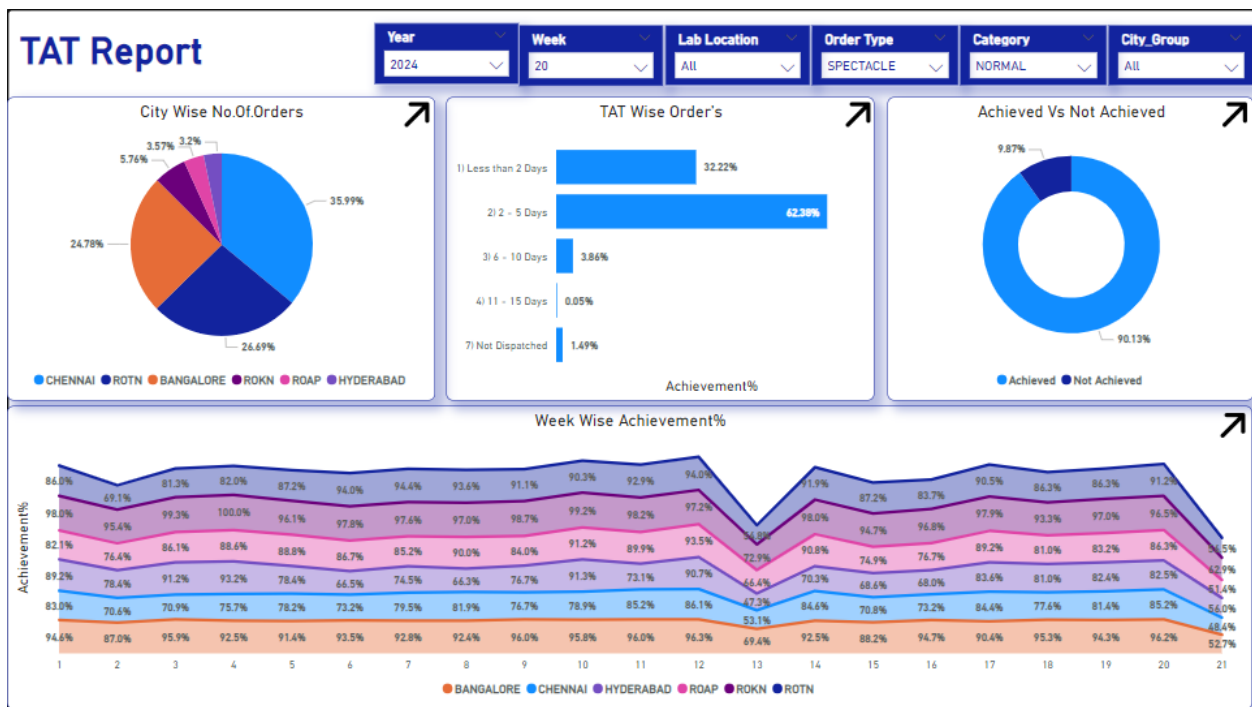


Figure 07. Overview of the Power BI dashboard

The Power BI dashboard provides a comprehensive TAT (Turnaround Time) Report, including several visualizations and metrics to evaluate performance across different cities and categories. Here's a detailed breakdown of each section:



Figure 08. Overview of the Filters

Filters

- Year
- Week
- Lab Location
- Order Type
- Category
- City Group

Visualizations and Metrics

City Wise No. of Orders (Pie Chart)

- Chennai
- ROTN (Rest of Tamil Nadu)
- Bangalore
- ROKN (Rest of Karnataka)
- ROAP (Rest of Andhra Pradesh)
- Hyderabad

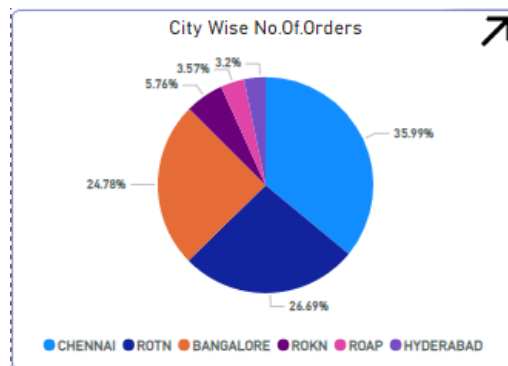


Figure 09. City Wise No. of Orders (Pie Chart)

The color coding for each city segment enhances the clarity and differentiation, making it easy to compare the order volumes at a glance. This visualization effectively emphasizes the concentration of orders in key regions and highlights potential areas for capacity planning and resource allocation.

TAT Wise Orders (Bar Chart)

- Less than 2 Days
- 2 - 5 Days
- 6 - 10 Days
- 11 - 15 Days
- 15 - 20 Days
- More than 20 Days
- Not Dispatched

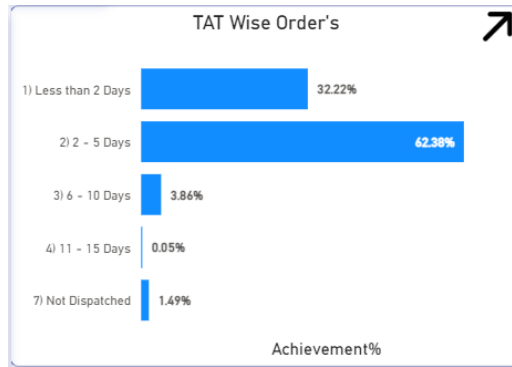


Figure 10. TAT Wise Orders (Bar Chart)

The chart effectively highlights the efficiency and speed of order processing, with most orders falling within acceptable TAT ranges, and helps identify areas needing attention for further improvement.

Achieved vs Not Achieved (Donut Chart)

- Achieved
- Not Achieved

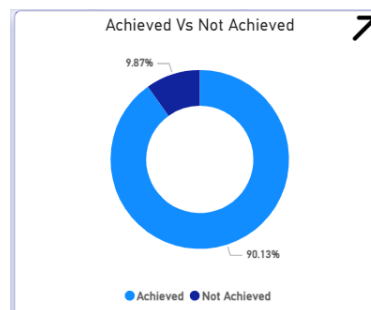


Figure 11. Achieved vs Not Achieved (Donut Chart)

The chart clearly shows that the majority of the instances fall into the "Achieved" category, significantly outnumbering the "Not Achieved" category. This visual representation effectively highlights the overall success rate.

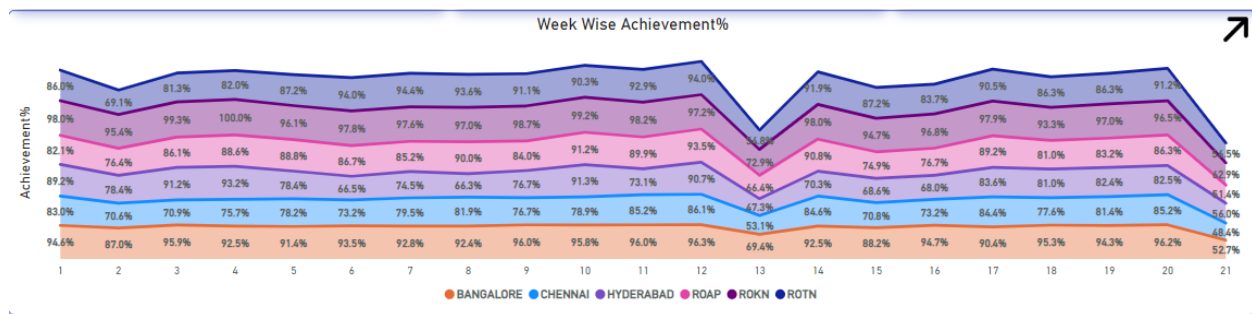


Figure 12. Week Wise Achievement% (Stacked Area Chart)

Displays achievement percentages for different cities, including Bangalore, Chennai, Hyderabad, ROAP, ROKN, and ROTN.

Achievement percentages indicate weekly variations in performance across different cities.

The other two Power BI reports visualizes weekly customer order metrics, showing total orders, billed and unbilled orders, processing times (ranging from less than two days to more than twenty days), fitment status, Dispatching status, orders achieved by TAT, and the achievement percentage, providing a comprehensive overview of order processing efficiency and performance.

5.Insights

This report provides an in-depth analysis of the performance metrics for spectacle orders, focusing on various aspects such as geographic distribution, turnaround time (TAT), achievement rates, and weekly trends. This is a product-based TAT report designed to help stakeholders understand key performance indicators and identify areas for improvement to ensure timely deliveries.

Performance Distribution: The majority of orders are concentrated in Bangalore (35.99%), followed by ROTN and Chennai.

TAT Performance: Most orders fall within the 2-5 days TAT bracket, with a substantial number also being completed in less than 2 days.

Achievement Rates: A high achievement rate of 90.13% shows that most orders are completed within the desired TAT.

Weekly Trends: The achievement rates across cities show fluctuations week by week.

This dashboard effectively tracks and visualizes the performance metrics for spectacle orders, allowing stakeholders to identify areas for improvement and ensure timely deliveries.

6.Discussion

The implementation of Turnaround Time (TAT) analytics using Power BI for Specs-makers provided a comprehensive analysis of the job order lifecycle, spanning from initiation to final dispatch. With over 250 stores across five states, Specs-makers needed to streamline their processes and enhance operational

efficiency. This project aimed to identify bottlenecks, optimize workflows, and enable data-driven decision-making to improve customer satisfaction and profitability.

The project involved integrating data from the Optic Pro application, which was extracted in CSV format, and utilizing key tables such as Store Master, Sales Register, Date Table, and Lens Master. We developed DAX (Data Analysis Expressions) queries to compute essential metrics and create interactive visualizations. The design of the Power BI dashboard focused on user-friendly navigation, incorporating tabs, filters, and intuitive layouts to ensure that stakeholders could easily interpret the data.

Key metrics such as order processing times, stock management efficiency, vendor performance, and fitting processes were highlighted in the analysis. Specific calculations included Job Order Raised to Fitment In-House, and Job Order Raised to Item Dispatched to Outlet, categorized into various timeframes for performance assessment. The visualizations provided insights into historical trends, business performance, and critical operational metrics, allowing for quick identification of areas needing improvement.

The analytical insights gained from this project revealed significant performance variations among different stores and regions, helping to identify both high-performing areas and those requiring intervention. The inclusion of "what-if" analysis enabled predictive planning, allowing the company to simulate various scenarios and prepare accordingly. The use of universal design principles ensured that the dashboard was accessible and easily navigable by all users, regardless of their technical expertise.

7.Conclusion

The Power BI TAT analytics project for Specsmaakers successfully demonstrated the potential of data-driven insights to optimize retail operations. By integrating and analyzing data from multiple sources, we created a comprehensive dashboard that provided clear and actionable insights into the job order lifecycle. The project achieved its objectives by providing a detailed analysis of TAT metrics across various stages, highlighting operational inefficiencies and bottlenecks, enabling data-driven decision-making and strategic planning, and delivering a user-friendly and accessible dashboard for stakeholders.

In conclusion, the application of Power BI for TAT analytics has significantly enhanced Specsmaakers' ability to monitor, analyze, and improve their operational processes. This project has laid the foundation for ongoing improvements, ensuring that Specsmaakers can continue to meet customer expectations and achieve sustainable growth in a competitive market. Through the insights and tools developed in this project, Specsmaakers is well-equipped to optimize their operations, enhance customer satisfaction, and drive profitable growth.